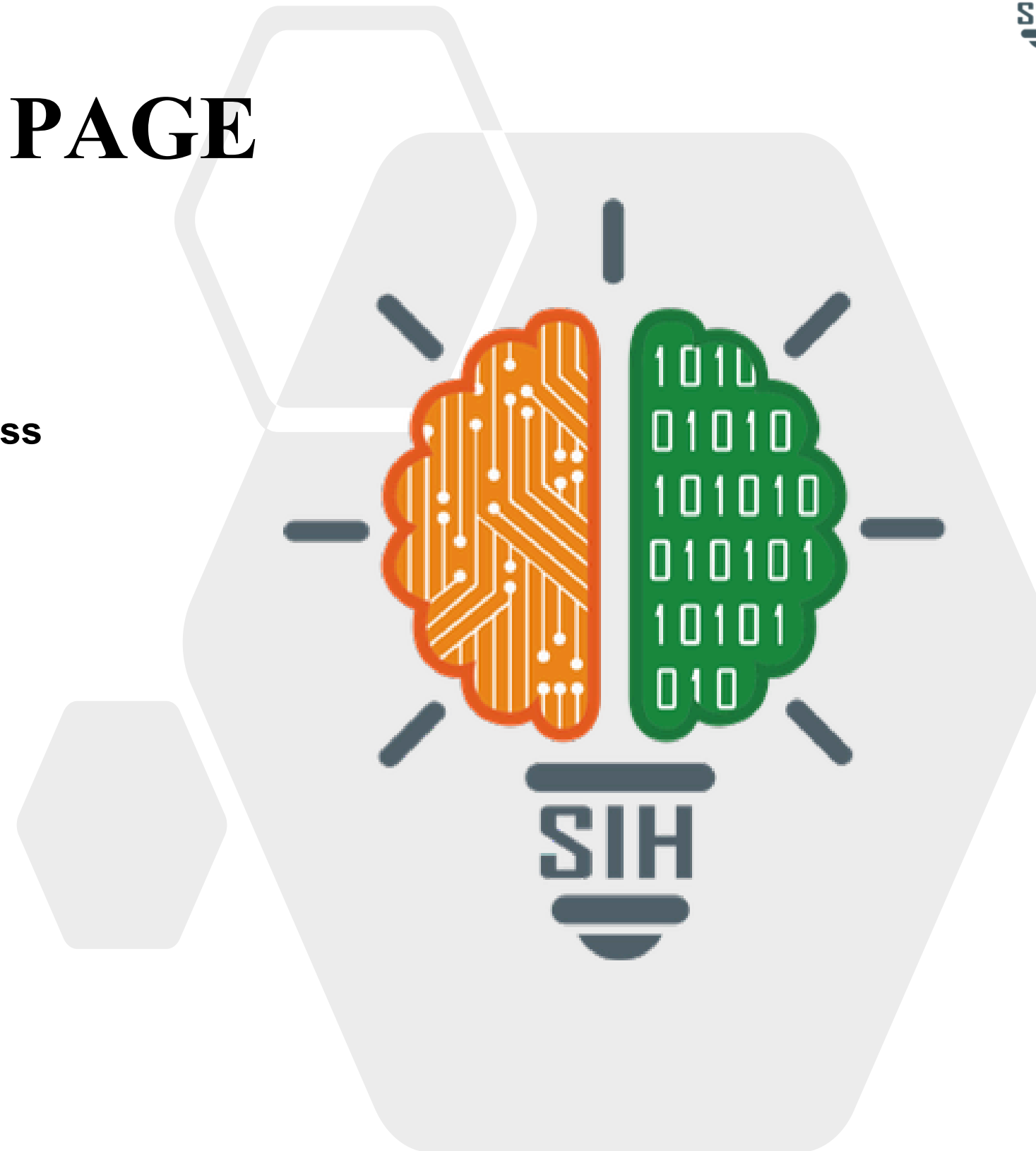


TITLE PAGE

- Problem Statement ID- 26186
- Problem Statement Title- AI-Based Predictive Personnel Stress and Welfare Monitoring System for Uniformed Forces
- Theme- MedTech / BioTech / HealthTech
- PS Category- Software
- Team ID- CSE-SIH-038
- Team Name (Registered on portal)- PsychoLogics



- **Proposed Solution: A Privacy first Early Warning Welfare Monitoring system for Uniformed Personnel**
- Fuses voluntary wellness assessments with HR, deployment, leave, workload data to provide contextualized welfare-risk assessment rather than relying on isolated indicators.
- Need to know RBAC: Personnel access their own wellness data; Welfare Officers access detailed information for assigned cases; Commanders receive appropriate unit-level/aggregated trends.
- Trajectory based risk detection and explainable intervention: tracking changes in wellbeing and organizational conditions over time, identifying worsening trajectories, explaining contributing indicators, and trigger appropriate human welfare review/contact.

Unique Value Proposition

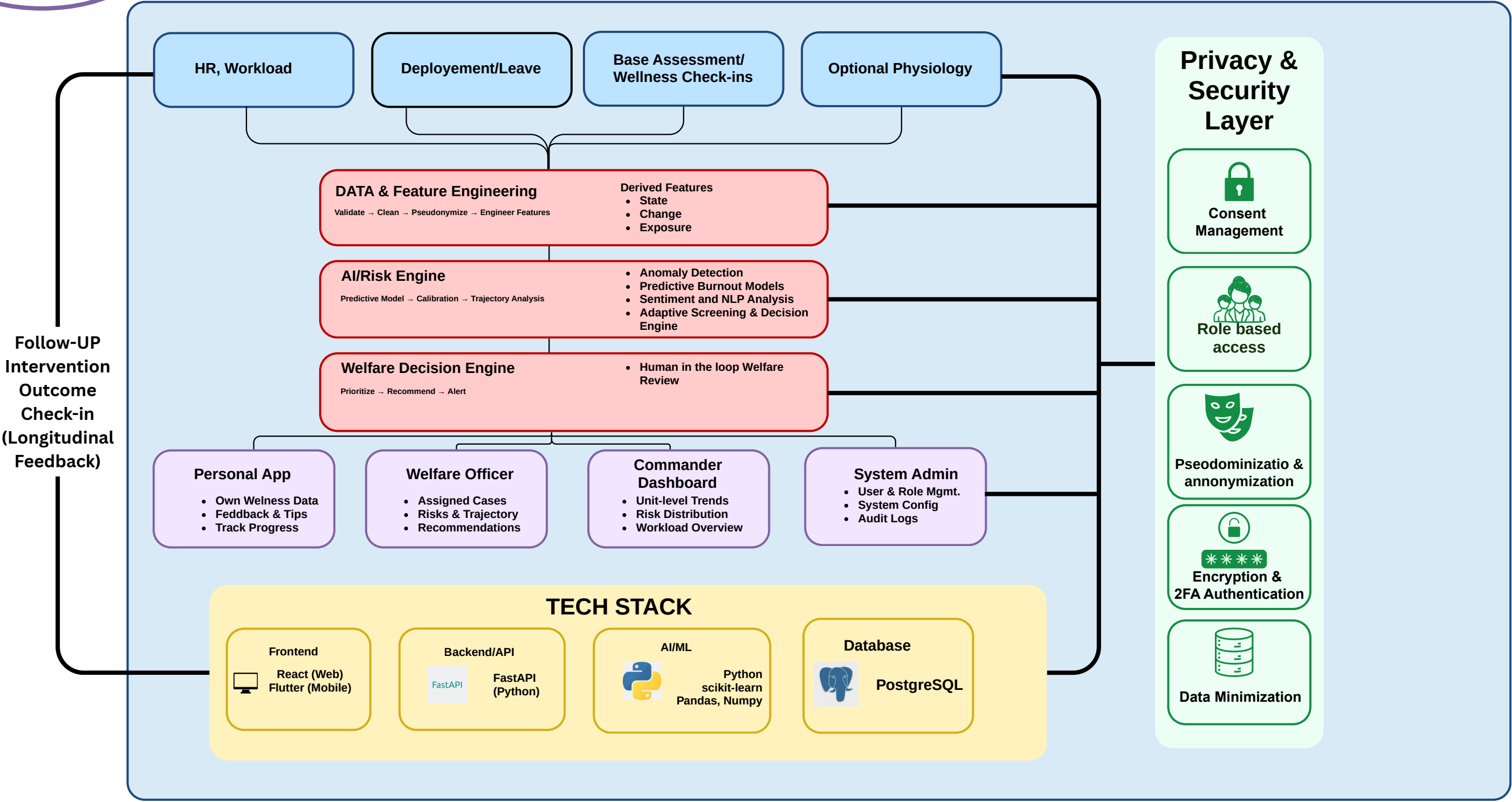
- **Longitudinal & Explainable Risk Detection** to Track changes over time, identify worsening trajectories, surface contributing indicators and trigger appropriate human welfare review.
- Psychology-informed assessment: Using validated psychological constructs/scales to contextualize voluntary self-reported wellbeing data.
- Calibrated & explainable risk
- Provides confidence and contributing indicators to support human review rather than opaque predictions.

How it addresses the problem

Shifting welfare management from reactive, manual self-reporting to proactive intervention.

TECHNICAL APPROACH

WMS - Technical Architecture



Feasability	Potential challenges and risks	Stratergies for Overcoming the Challenges
TECHNICAL FEASIBILITY - Cloud integration - Use of already available technologies OPERATIONAL FEASIBILITY - Designed as a welfare-support and early-warning system rather than a disciplinary tool - Scalable and adaptable for large-scale deployment PSYCHOLOGICAL FEASIBILITY - ICD- and DSM-informed - Use of established ethical principles	Cybersecurity Privacy of sensitive personnel data risk- breaches can cause theft and losses Data Gathering Lack of real military datasets risk- difficulty in model training Fairness Testing AI bias risk- incorrect correlations Adoption Barrier Personnel may not trust the system risk- gathering of incomplete data	- explainable AI - Encryption, Role-Based Access Control - Data Minimization - a Fairness Monitoring Module - Synthetic Dataset - Using wearables for data collection for personnel for biometric data.

Potential Impact on Personnel	Benefits of the Solution
1. Earlier recognition of psychological distress. 2. Assessment tailored to individual responses. 3.Encourages help-seeking and self-awareness. 4. Better connection between veterans and support services. 5. Supports long-term psychological wellbeing. 6.May improve functioning in family and social life. 7.Creates a preventive approach to veteran welfare.	1.Adaptive AI reduces unnecessary questions. 2.Faster and more efficient screening. 3. Converts responses into actionable welfare insights. 4.Helps professionals prioritize cases requiring attention. 5. Potential reduction in long-term welfare costs. 6. Scalable to large veteran populations. 7.Enables structured follow-up over time.

- **Datasets referenced for method design (not CRPF operational data)**
- •WESAD (Wearable Stress and Affect Detection) – Schmidt et al., ACM ICMI 2018 – physiological stress dataset
- •DAIC-WOZ (Distress Analysis Interview Corpus) – USC Institute for Creative Technologies – behavioural distress-screening corpus
- **Key methods**
- •SHAP – Lundberg & Lee, "A Unified Approach to Interpreting Model Predictions," NeurIPS 2017 – explainability layer
- •Comprehensive Soldier and Family Fitness – US DoD resilience/wellness program – precedent for military welfare programs
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